Mini NPU Compiler Design Spec Template
Lecture 20 Capstone

Use this document as the final submission template for the capstone design review.

1. Problem Scope

- **Workload and model block: ________________________________**
- **Input IR and export path: ________________________________**
- **Supported shapes, layouts, and dtypes: ________________________________**
- **Explicitly unsupported cases: ________________________________**
- **Fallback or reject policy: ________________________________**

2. Target NPU Assumptions

- **Native tile shape: ________________________________**
- **SRAM / ACCUM capacity: ________________________________**
- **DMA alignment and bandwidth assumptions: ________________________________**
- **Quantization support: ________________________________**
- **Runtime/firmware constraints: ________________________________**

3. IR Stack and Dialect Contracts

- **Input dialects: ________________________________**
- **npu_graph responsibility: ________________________________**
- **npu_kernel responsibility: ________________________________**
- **npu_rt responsibility: ________________________________**
- **Full conversion boundaries: ________________________________**

4. Legal Op Set Matrix

Layer	Legal ops	Shape/layout/dtype constraints	Unsupported policy
Input	StableHLO/TOSA/Linalg subset		
Graph	npu_graph.matmul_epilogue, requant, layout		
Kernel	npu.matmul, npu.dma, npu.barrier		
Runtime	npu_rt.descriptor, command, launch		

5. Pass Pipeline

Pass	Anchor	Input -> output	Analysis	Failure/test
npu-decompose-composite	builtin.module			
npu-legalize-input-to-graph	func.func			
npu-fuse-matmul-epilogue	func.func			

npu-apply-schedule	func.func			
one-shot-bufferize	builtin.module			
npu-lower-to-kernel	func.func			
npu-lower-to-runtime	builtin.module			

6. Schedule, Memory, and Quantization

- **Tile shape and loop order:** ______________________________
- **SRAM/ACCUM footprint calculation:** ______________________________
- **DMA order and barrier placement:** ______________________________
- **Layout propagation and materialization points:** ______________________________
- **Accumulator width and requantization formula:** ______________________________
- **INT4/INT8 packing policy:** ______________________________

7. Runtime ABI

Artifact	Fields	Versioning/check	Debug/replay
Tensor descriptor	address, shape, stride, dtype, layout, quant		
Command buffer	header, descriptor table, records, relocation		
Manifest	artifact files, target, ABI, checksum		
Runtime API	create, submit, wait, destroy		

8. Validation and Evaluation

- **Positive FileCheck test:** ______________________________
- **Negative diagnostics:** ______________________________
- **Numerical golden reference:** ______________________________
- **Performance model assumptions:** ______________________________
- **Runtime schema/replay test:** ______________________________
- **Known risks and future work:** ______________________________